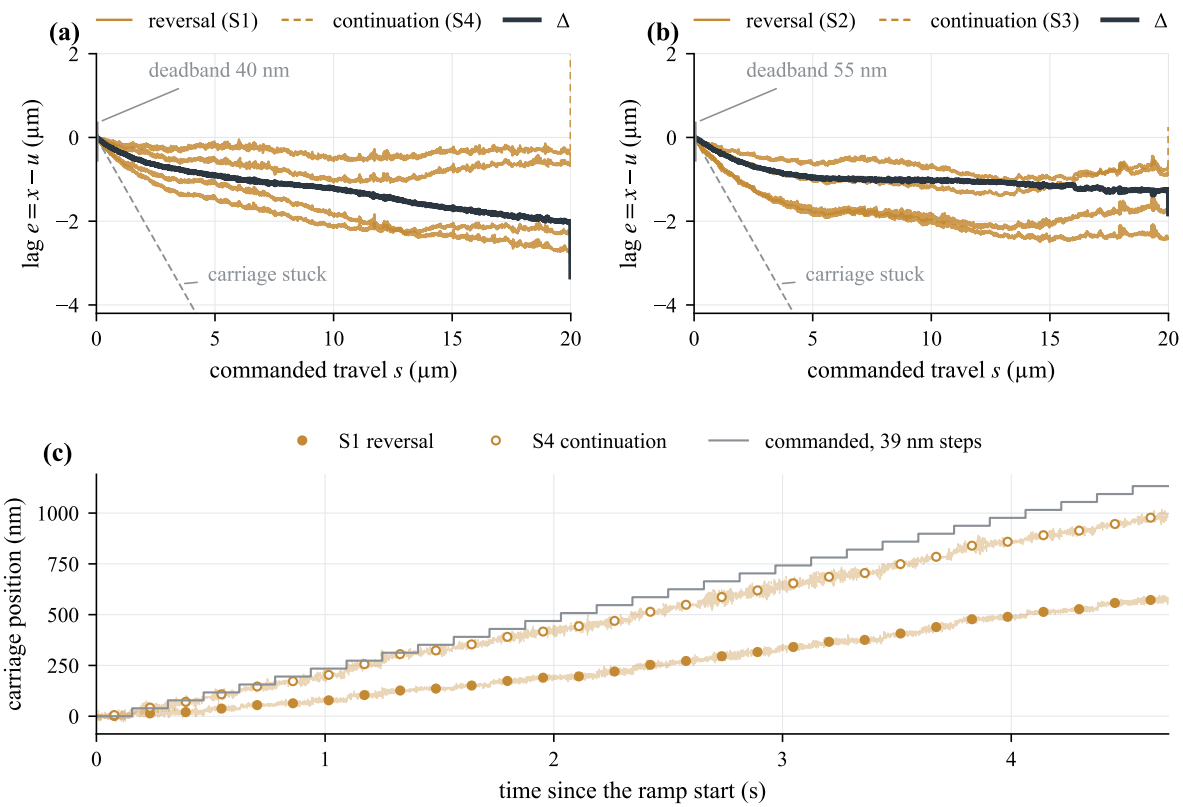

(a)
reversal (S1)
continuation (S4)
Δ
deadband 40 nm
carriage stuck
lag $e = x - u$ (µm)
commanded travel s (µm)
(b)
reversal (S2)
continuation (S3)
Δ
deadband 55 nm
carriage stuck
lag $e = x - u$ (µm)
commanded travel s (µm)
(c)
S1 reversal
S4 continuation
commanded, 39 nm steps
carriage position (nm)
time since the ramp start (s)